

December 2024

B. Tech. (ME) (Fifth Semester)

**Solid and Hazardous Waste Management
(OEC-ME-509)**

Time : 3 Hours]

[Maximum Marks : 75

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other. Draw neat and clean diagrams wherever necessary.

Part A

1. Write short notes on each of the following :

- | | |
|-----------------------|-----|
| (a) Radioactivity | 1.5 |
| (b) Sanitary landfill | 1.5 |
| (c) Vermicomposting | 1.5 |

(d)	Incineration	1.5
(e)	In-situ Remediation	1.5
(f)	Waste reduction	1.5
(g)	C/N ratio	1.5
(h)	Recycle verses Reuse	1.5
(i)	Air stripping	1.5
(j)	Pyrolysis.	1.5

Part B

2. (a) What is an engineered landfill ? With the aid of a neat sketch, explain the design landfill for solid and hazardous wastes. **10**
- (b) Explain Stabilization and solidification of hazardous waste. **5**
3. (a) Discuss the salient features of Solid Waste Management Rules, 2016. **5**
- (b) Answer the following : **2×5=10**
 - (i) Explain the various means used for transportation of solid waste.

- (ii) Explain the objectives and various processing techniques used for MSW disposal.

4. Answer the following : $2 \times 7.5 = 15$

- (i) What is risk and environmental risk ? Explain methods of risk assessment.
- (ii) Explain various methods of waste minimization and resource recovery.

5. (a) What is a bio-reactor ? Discuss the anerobic decomposition of soild waste. 5

(b) Answer the following : $2 \times 5 = 10$

- (i) Explain the storage and handling of hazardous waste.
- (ii) What are radioactive wastes ? What are its effects on human and environment ?

6. (a) Describe the various physciochemical processes used for the treatment of hazardous waste. 10

(b) Explain the principles of biodegradation of toxic waste. **5**

7. Discuss the rules associated with biomedical waste handling and describe the colour coding of bins used for biomedical waste storage. **15**